

Subham Dey

Curriculum Vitae

Education

- Nov. 2019– July 2023 **Ph.D. (Control Systems)**, LAMIH (CNRS) (UPHF) (France) and University of L'Aquila (DISIM, DEWS) (Italy), (Pr. Stefano Di Gennaro, Pr. Mohamed Djemai (CNU 61) and Pr. Michael Defoort (CNU 61)).
- May 2017– May 2019 **Master of Electrical Engineering**, BIT-Mesra (Course Work) (IIT-Kanpur (Thesis Work)), (India), (First Class with Distinction).
Specialized in Control Systems
- May 2012– May 2016 **Bachelor of Electrical Engineering**, Jadavpur University, Kolkata, (First Class).
Specialized in Instrumentation and Control

Postdoc research

- Title *Personalized optimized brain stimulation, systems neuroscience and computational biology.*
- Section (Noninvasive Neuromodulation Unit of NIMH. (17 Jun 2024– 1 May 2026) on lien to National Library of Medicine (NLM) (since 1 May 2026), University of Texas at Austin (Sep 2023 - Jun 2024))
- Description
- Edge based controllability analysis of monkey structural connectome in order to find optimal sites of invasive and noninvasive brain stimulation. Using controllability metrics to find a therapeutic planning mechanism for patients subjected to ketamine/ECT treatment. Communication of findings at a conference (IEEE EMBS 2025). Talk and poster presentation at SOBP 2026.
 - Direct data-driven model predictive control for TMS–EEG closed–loop brain stimulation experiments.
 - Evaluating the attractor dynamics from resting state functional MRI datasets and trying to know if brain stimulation modalities change the attractor landscape in case of active stimulation whereas sham stimulation has no effect on it. Population-level analysis of multielectrode neural data collected simultaneously from multiple brain areas when monkeys were trained to perform an attention-based learning task (Individual data collaboration).
 - Mathematical modeling of genome evolution.

Ph.D. Thesis

- Title *Contributions on event-triggering control for systems on arbitrary time domains.*

Supervisors Pr. Mohamed Djemai (LAMIH), Pr. Michael Defoort (LAMIH), Pr. Stefano Di Gennaro (University of L'Aquila)

Description

- To analyze the observability of switched systems on non-uniform time domain using measure chain theory.
- To design observers for switched systems on non-uniform time domain using measure chain theory.
- Event-triggered observer design for systems using measure chain theory.
- To apply these techniques to networked control systems.

Masters Thesis

Title *Non-linear Non-singular Terminal Sliding Mode attitude control of satellites.*

Supervisors Dr. Dipak Kumar Giri (IIT-Kanpur) & Dr. Vijaya Laxmi (BIT-Mesra)

Description This thesis aims at exploring the problem of attitude control of satellites actuated by magnetic as well as magneto-Coulombic actuators resulting in a combined actuation scenario and designing controllers that are robust, fault-tolerant and also ensures finite time stability.

Bachelors Thesis

Title *Sliding mode control of 5-DOF active magnetic bearing systems.*

Supervisors Dr. Amitava Chatterjee

Description Implemented integral sliding mode control for active magnetic bearing systems on Simulink and tried to extend to other nonlinear systems like blood glucose regulation.

Experience

Research

2019-2022 **Research Assistant**, BITS-PILANI (INDIA), IISC-BANGALORE (INDIA) and CNRS FRANCE.

- Working on topics related to coalitional game theory.
- Working on distributed optimization and control theory.
- Working on topics related to formal analysis.

2016-2017 **Project Fellow**, CSIR-INSTITUTE OF GENOMICS AND INTEGRATIVE BIOLOGY, New-Delhi.

Detailed achievements:

- Learned basics of molecular biology and wet lab skills.
- Designed a mathematical model for a gene caliper, which indirectly measures the CAG repeat length as a measure of fluorescence from a reporter gene.
- Data analysis of confocal images of primary neuronal cultured cells to quantify neurite length and arborization.

2016 **Summer Intern**, IISER-KOLKATA, Kolkata.

Analysis of human brain functional network from fMRI data when the stimulus is a multisensory one using model-based approach (GLM) and Independent Component Analysis (ICA).

2014-2015 **Research Fellow**, JADAVPUR UNIVERSITY, Kolkata.
Fractal analysis of EEG data obtained from electrodes in the frontal lobe of brain.

Awards

- 2016 National Network for Mathematical and Computational Biology (NNMCB) Summer Internship. (IISc Bangalore) (Award: 2015-16 academic year)
- 2016 GATE 2016 (MHRD-Govt. of India): A postgraduate test for financial assistance for masters study in Electrical Engineering. (95 percentile)
- 2024 Best poster award in the Biocontrol workshop 2024 (sponsored by IFAC Activity Fund)
- 2025 NeuroXAI: Explainable AI for Understanding Biological Brains, workshop organizer (34th Annual Computational Neuroscience Meeting, Florence, Italy)

Skills

Packages FIJI, Cell Profiler, NEURON, SPM
Software MATLAB, $\text{\LaTeX}$, Python, C, Julia
Wet Lab PCR, Restriction Cloning, Transformation, electrophysiology

Publications

Journal Publications:

F.B. Bisso(*), S. R. Lozano (*), S. Dey (*), G. B. Stan, and C. C. Samaniego "A Reference-Free Molecular Feedback Controller for Inducing Sustained Oscillations in Gene Regulatory Networks ", (IEEE Control Systems Letters) (Accepted in press) (*: Equal contributions) (Also at CDC 2026)

S. Dey et al. "Safety-critical event-triggered control for quasi-linear systems on measure chains ", IEEE Control Systems Letters, Vol. 8, 1042-1047. (Also at CDC 2024)

S. Dey et al. "Event-Triggered Control for Systems on Non-Uniform Time Domains Using Measure Chain Theory ", IEEE Transactions on Circuits and Systems II: Express Briefs. 2021 Jun 29;69(3):1198-202.

S. Dey et al., "Observer based Leader-Follower Bipartite Consensus with Intermittent Communications using Lyapunov Functions and Time Scale Theory", IEEE Control Systems Letters, Vol. 5, Issue 6, December 2021, 1904-1909. (Also at ACC 2021)

S. Dey, D.K. Giri, K. Gaurav and V. Laxmi "Robust Non-singular Terminal Sliding Mode Attitude Control of Satellites", ASCE-Journal of Aerospace Engineering, Vol. 34, Issue 1, January 2021

A.K. Maity, R. Pratiher, A. Mitra, S. Dey et. al. "Multifractal detrended fluctuation analysis of alpha and theta EEG rhythms with musical stimuli", Chaos, Solitons and Fractals, Volume 81, Part A, December 2015, Pages 52-67.

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S. Dey, "Edge controllability is associated with treatment response to repetitive transcranial magnetic stimulation in depression." (Under review)

References

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Pr. Michael Defoort (Michael.Defoort@uphf.fr) (Universite Polytechnique Hauts-de-France (Le Mont Houy), INSA, LAMIH (UMR CNRS 8201), 59313 Valenciennes, France)
Dr. Sarah Hollingsworth Lisanby (holly.lisanby@asu.edu) (Arizona State University)
Dr. Bruce Luber (Retired) (bruce.luber@nih.gov) (Noninvasive Neuromodulation Unit, NIMH)

Reviewing activity

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Journal of Franklin Institute, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Aerospace and Electronic Systems, IEEE Transactions on Control of Network Systems, IEEE Control Systems Letters, IEEE Transactions on Circuits and Systems I: Regular Papers, Nonlinear Dynamics, IEEE Transactions on Medical Robotics and Bionics, Frontiers of Information Technology & Electronic Engineering, IEEE Transactions on Cybernetics, IEEE Transactions on Industrial Informatics, Systems & Control Letters, European Journal of Control, ISA Transactions, Proceedings of the Institution of Mechanical Engineers. Part I: Journal of Systems and Control Engineering, Systems & Control Letters, Psychiatric Research Neuroimaging, Multiple Conferences (CODIT(2023,2024,2025,2026), ECC(2022,2023,2024,2025), CDC(2023,2024,2025))